

Sound

Every sound starts with a vibration, like the quivering of a guitar string. The vibration squeezes and stretches the air, sending its energy out in waves in all directions. This is a sound wave.

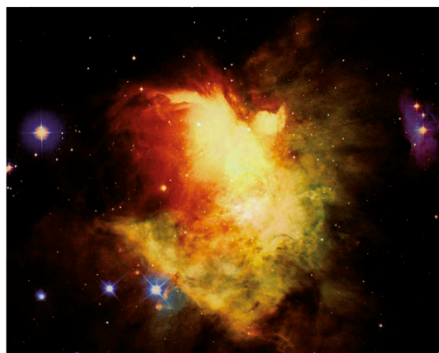


Sound notes

When you blow across a bottle, the air inside vibrates. Small air spaces vibrate more quickly than large spaces, making higher notes. So partly empty bottles produce lower notes than fuller ones.

Silent space

Sound can travel through solids, liquids, and gases, but it can't travel where there is no matter. There is no sound in space because there is no air.



Sound waves travel through air like a wave along a coiled spring.

How hearing works

When a sound reaches your ears, it makes your eardrums vibrate. The vibrations are passed to your inner ear through tiny bones. From here, nerves send messages to your brain that allow you to recognize the sound.

Measuring sound

Loudness is measured in decibels.



Leaves rustling nearby make a sound of only 10 decibels.



Somebody **whispering** close by measures about 20 decibels.



City traffic reaches approximately 85 decibels.



Drums being played nearby makes a sound of around 105 decibels.



Road-drills measure about 110 decibels from a close distance.



A **lion's roar** would measure 114 decibels if you were close enough.



Fireworks can measure 120 decibels or more.



The sound of **jet engines** sometimes hit 140 decibels if heard from nearby.